FILALOOM: A TOOL CHANGING SYSTEM FOR AUTOMATIC YARN-EMBEDDING IN FDM-PRINTING

A PREPRINT

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ABSTRACT

This work showcases FilaLoom, a tool changer that enables embedding yarns directly into Fused Deposition Modeling (FDM) 3D printing. FilaLoom allows FDM-printers to switch between printing and yarn-embedding operations, enabling the automatic 'weaving' of yarns into printed objects in a multi-material fabrication process. FilaLoom is a DIY, easily implemented add-on that empowers designers and researchers with a new digital fabrication tool for printing objects with increased textileness. We explore the design space and develop several printing strategies that demonstrate how FilaLoom enables the fabrication of FDM-prints with textile-like woven structures, soft yarn textures and cable-driven interactive properties.

Keywords FDM-printing · Textiles · Multi-material printing · Tool changer · 3D weaving

1 Introduction

Fused Deposition Modeling (FDM) has become a widely available fabrication technology, enabling the rapid and low-cost production of physical objects from digital designs. An important characteristic of FDM is its adaptability as an experimental fabrication platform. Makers and researchers can modify materials, software, toolpaths, and hardware to explore fabrication processes beyond conventional material extrusion. Such modifications include altering material compositions before or during extrusion [1, 2, 3], developing custom toolpaths and process parameters to achieve particular functional or aesthetic characteristics [4, 5, 6], and adapting printer hardware to introduce additional fabrication capabilities [7, 8, 9].

These custom approaches include broadening the material palette by embedding non-printable materials into prints to extend their functional and aesthetic properties [10]. This has resulted in FDM-printed objects with, for example, increased strength & durability [10], electronic capabilities [11, 12] and shape-changing properties [13, 14, 15].

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Figure 1: FilaLoom system and fabrication outputs: (a) vertical woven objects, (b) planar semi-woven textiles, (c) FilaLoom attached to a Bambu Lab A1 FDM-printer, (d) soft yarn textures embedded into printed objects, and (e) a cable-driven textile.

Yarn stands out as a promising candidate among these non-printable materials. Yarns exhibit unique malleability and strength, as well as characteristic aesthetic and tactile properties that give textiles their distinct feel, known as "textileness" [16, 17]. This makes yarn well-suited for fabricating objects with soft and textile-like properties. Prior work demonstrates the potential of combining fibrous materials and textile processes with 3D printing. Some approaches extend printers with hardware for automatically depositing or embedding fibers [7, 8], while others combine printed structures with manual weaving or needlecraft [18, 19]. Together, these works show how fibers and additive fabrication can be combined through both automated and craft-based processes to create objects with new material, structural, and tactile qualities. Building on this potential, we introduce *FilaLoom*, which explores a design space in which continuous yarn is automatically placed within FDM-printed objects to create woven structures, soft surface textures, and cable-driven actuation.

FilaLoom (Fig. 1) is a mechanical tool-changing add-on that enables desktop FDM-printers to alternate between filament deposition and yarn-placement operations. It can be attached and detached without permanent printer modification and operates through the printer's existing motion system, requiring no additional electronics. In summary, our work contributes the following:

- An accessible hardware add-on that enables consumer-grade FDM-printers to automatically embed yarns in printed objects in a one-step digital fabrication process.
- A characterization of the fundamental design space of yarn-embedded printing with FilaLoom, together with practical guidelines for its use.
- Five yarn-embedding print strategies for fabricating FDM-printed objects with woven structures, soft textures, and cable-driven actuation.
- A series of fabricated examples demonstrating FilaLoom's potential across a range of applications, including functional soft objects, decorative items, and interactive textiles.

By enabling automated yarn placement within FDM-printed objects, FilaLoom provides a platform for designers and researchers to explore a new design space at the intersection of additive fabrication and textile making, including novel material interactions, fabrication processes, and soft, textile-like structures.

2 Related Work

Various researchers have used both automatic and manual approaches to explore the use of 3D printed filaments to generate textile-like material outcomes, while more recently some have combined FDM-printing with yarns or fibers. This section outlines the current state of research into printing textile-like materials and embedding fibrous material into FDM-printing.

2.1 FDM-printed textiles and fibers

Simulating the properties and structure of textiles through FDM-printing alone has been widely explored (see [20] for an overview). Today, many researchers and designers print thin TPU lines flat on the print bed to produce a mesh-like sheet, relying on TPU's flexibility to create modular clothing and accessories [21, 22]. Other examples exploit controlled under-extrusion to create tulle-like textiles [23], print vertical pillar 'warps' and string filament through them as a 'weft' to produce woven structures [20], utilize a dual nozzle printer to fabricate woven structures from PLA and TPU with plain, twill, satin and jacquard patterns [24], and print alternating trapezoid zigzag layers to create 3D printed spacer fabrics [25].

Researchers have also attempted to print fiber itself. This is most commonly explored through exploiting FDM-printers' stringing 'defect' to produce hair-like textures [26], recreate images in a fiber-art style [27], and apply the 'suspended filament technique' to create structures that hang suspended from printed strings [28, 29].

While these techniques demonstrate the capabilities of FDM-printing to fabricate textile-like outcomes with considerable freedom in geometry and texture, they are limited to printable materials and therefore cannot replicate the unique tactile qualities found in traditional textiles.

2.2 Fiber and Yarn-embedded FDM-printing

Embedding yarn or fiber into FDM-printing has been explored in several different contexts in both academia and industry. Rhapso [8] is a custom add-on for FDM-printers consisting of a rotating ring with a yarn carrier attached. Custom software drives a stepper motor on the ring to automatically position the yarn in the XY plane, and the printer's GCode is modified to print over the yarn in a specific order. This approach is suitable for automatically clamping yarns in a 3D printed object, enabling fabrication of cable-driven mechanisms, hinges and elastic springs.

Rhapso uses premade yarn on a spool; however, it is also possible to fabricate "yarn" within the printing process. XStrings [30] uses a dual nozzle FDM-printer to extrude a nylon cable inside PLA parts. Combining this method with an algorithm to generate motion primitives allows complex cable-driven mechanisms to be fabricated in a single print.

In an industrial context, several companies have successfully brought fiber-embedding FDM-printers to market. The Mark Two [31] and FibreSeeker 3 [32] have continuous carbon fibers embedded in a resin or thermoplastic filament, which is extruded out of the nozzle and cut after each print layer to reinforce the strength of the printed part.

While these approaches demonstrate the viability of yarn- and fiber-embedded FDM-printing, their applications are primarily limited to improving strength or adding cable-driven motion to rigid objects.

2.3 Yarn-FDM hybrid textiles

Yarn-FDM hybrid textiles are FDM-printed structures that integrate yarn as a core material component, aiming for a textile-like experience.

Yarn-FDM hybrid textiles have primarily been researched within the context of handmade craft. In EscapeLoom, Deshpande et al. 2021 generated FDM-printed hand looms with novel affordances, resulting in textiles with printed warps and yarn wefts inserted by hand. Other examples similarly generate 3D printed substrates, in which yarn is then manually embedded through basket weaving [33] or needle punching [19]. Such manual approaches connect digital workflows to textile craft traditions, exploring how this combination can produce new artifacts and extend meaningful handwork practices [34, 35, 36]. In contrast, some researchers have explored the fabrication of similar hybrid textiles by manually inserting the yarn into FDM-printers and printing on them [37], exploring filament-yarn material compatibility and fabrication methods across a broad design space.

This body of work establishes the design space and potential of yarn-FDM hybrid textiles through manual yarn integration, either post-printing or within the printing process itself. While manual integration is often a deliberate choice that embodies the craft aspect of traditional textile practices, automated yarn-embedding enables faster iteration and new fabrication possibilities. With FilaLoom, we present a tool that enables automatic fabrication of yarn-FDM hybrid textiles by embedding yarn as a core material component in FDM-printed structures, combining the unique material properties of yarn with the geometric freedom of FDM-printing in a single fabrication process.

3 FilaLoom

FilaLoom enables the automated embedding of yarns into the FDM-printing process. The system consists of three components that can be attached onto desktop FDM-printers with minimal effort (Fig. 2). Our prototype was created

for the Bambu Lab A1, a bed-sling printer. However, the general set-up and workflow are adaptable to most desktop FDM-printers, both bed-sling and core-XY models. The mechanism includes parts FDM-printed from PETG filament, and three commonly-available standard parts: a blunt tip dispensing needle, a 1.5x10x15 mm compression spring, and several 8x2mm neodymium magnets.

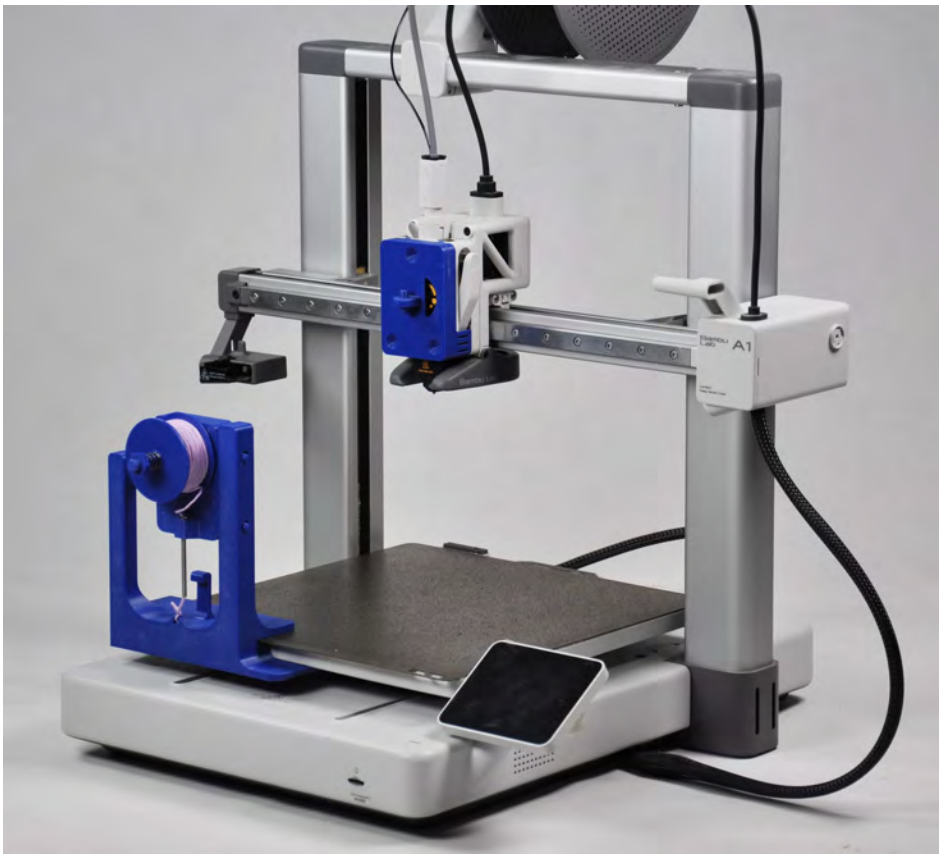


Figure 2: FilaLoom system attached to a Bambu Lab A1 FDM-printer.

3.1 Components

The yarn-embedding mechanism is a tool changer, allowing the printer to switch between printing and yarn-depositing operations by picking up and parking a yarn depositing tool (Fig. 3). The mechanism consists of three parts: a yarn depositor, a print head cover that can lock onto the yarn depositor, and a station at the edge of the print bed onto which the depositor can be parked.

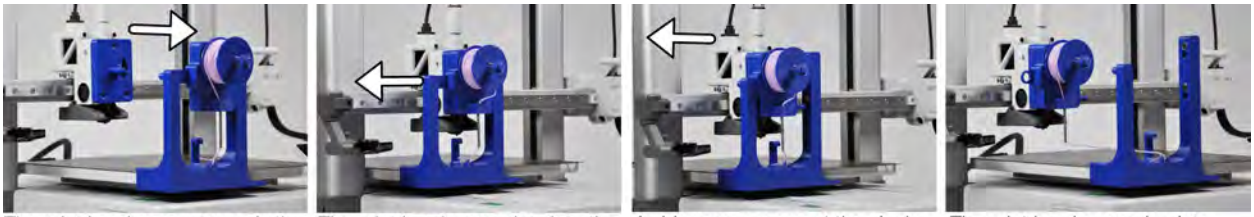


Figure 3: A FilaLoom-enabled printer picks up the yarn depositor, allowing it to weave yarn through a print.

3.1.1 Yarn Depositor

The yarn depositor (Fig. 4) is the component that embeds yarn into the print. It consists of a body with an axle protruding horizontally from it. A spool of yarn is mounted to this axle and locked in place with a washer, a compression spring and a bolt. This spring pushes the washer against the yarn spool, applying friction to the spool which results in yarn tension. The tension can be adjusted by rotating the bolt to tune the amount of friction between the washer and the spool. The yarn is guided from the spool through a blunt-tip hollow needle to precisely position the yarn in the print.

The yarn depositor also features two separate docking mounts: bars at the side to dock it onto the station, and an interlocking mechanism to lock it onto the print head cover. This interlocking mechanism is similar to those in the tool changers of the Prusa XL and Snapmaker U1 FDM-printers [38, 39]. It features a 'keyhole' with a slider lock and three cone-shaped protrusions for kinematic coupling.



Figure 4: FilaLoom's yarn depositor with constituent parts annotated, shown from the front (top) and back (bottom).

3.1.2 Print head cover

The print head cover (Fig. 5) features respective counterparts to the interlocking mechanism on the yarn depositor. A plus-shaped protrusion can be inserted into the 'keyhole' on the depositor. Three inverted cone holes pair with the cone protrusions on the yarn depositor to perform kinematic coupling, ensuring consistent positioning between print head and depositor. After insertion, the sideways motion the print head performs during a pick-up sequence moves the slider lock, tightly locking the yarn depositor onto the print head.

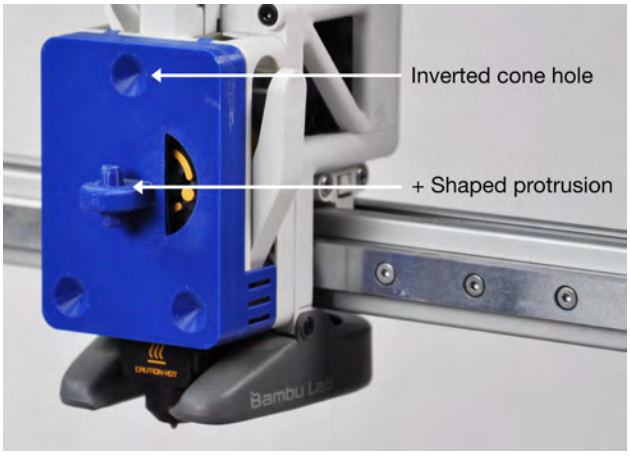


Figure 5: FilaLoom's print head cover with constituent parts annotated.

3.1.3 Station

To unlock the yarn depositor from the print head, it is parked onto the station (Fig. 6 left). The station is screwed onto the print bed at a fixed position. The station features two vertical pillars. The yarn depositor is docked onto the larger of

the two by slotting the bars on its side into corresponding holes on this pillar. Magnets in both the station and yarn depositor supplement this connection, helping to align the connection points and holding the depositor firmly in place. The second pillar features a protrusion that slots into a loop on the slider lock, so that the sideways motion used to dock the depositor onto the station simultaneously pulls the slider lock open, unlocking the depositor from the print head. The reverse motion pushes the slider lock in, locking the depositor onto the print head. The station also features a small loop to which the yarn is tied at the start of a print, serving as the initial attachment point from which the yarn is first pulled, and a hook that acts as an anchor point to maintain yarn tension between weaving sequences.

The station is also available in a double configuration that accommodates two yarn depositors (Fig. 6 right), enabling multiple yarns to be embedded within the same print.

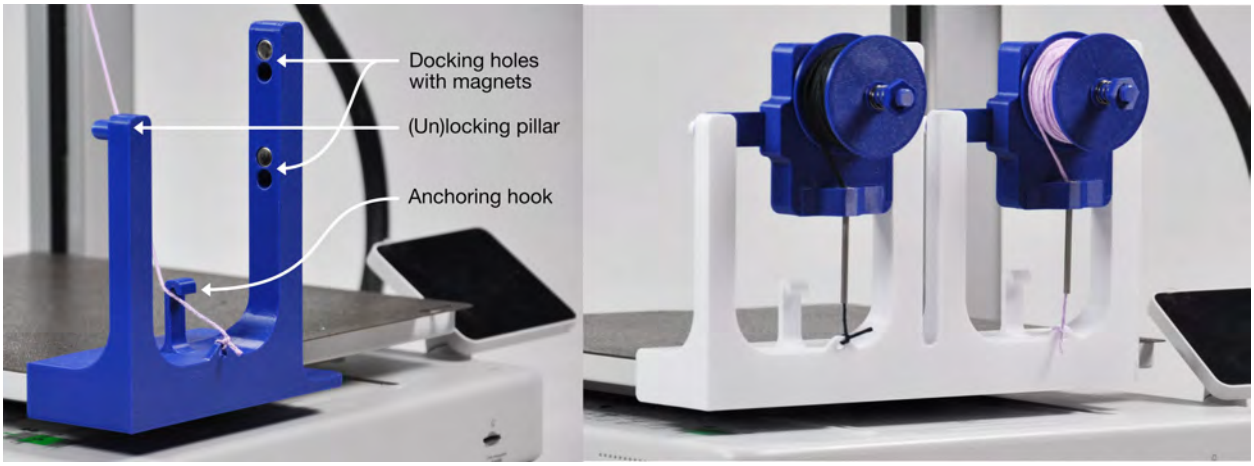


Figure 6: FilaLoom station configurations: **left**: Single station module with constituent parts annotated; **right**: Double station module featuring two yarn depositors parked side by side.

3.2 Working principle

To embed yarns within a print, FilaLoom relies on anchors, stubby vertical pillars with a crease or hook that hold the yarn in place and decide its position within the print. These anchors can be embedded within the printed part or an external sacrificial part. The working principle (Fig. 7) is as follows: The printer initiates the print and starts to build up the printed structure. Along with the printed structure, anchors are printed. The printer then initiates a 'pick-up' sequence, attaching the yarn depositor to the print head. As the print head moves, the yarn gets pulled away from its attachment point on the station, unwinding it from the spool. Using custom GCode, the movement of the print head is then controlled to pass the yarn through the printed structure from anchor to anchor. After weaving the yarn through the anchor points, the printer initiates a parking sequence, placing the depositor back onto the station. This frees up the print head to print subsequent layers, printing over the yarn and embedding it into the print. This print $\rightarrow$ pick-up $\rightarrow$ weave $\rightarrow$ park sequence then repeats until the print is finished.

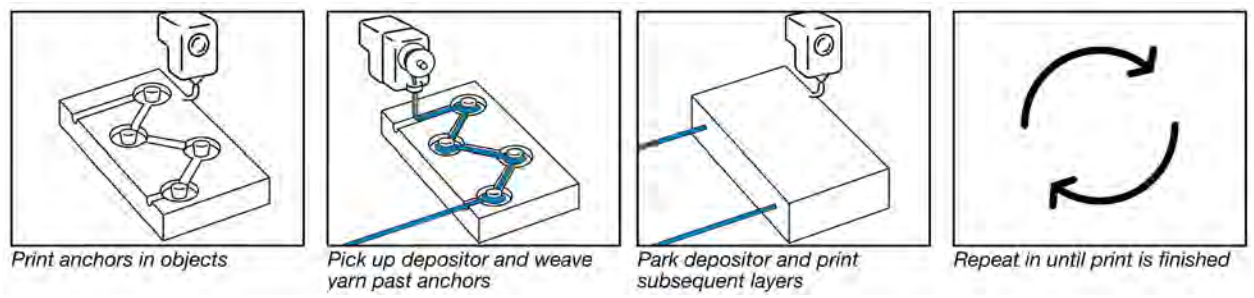


Figure 7: An illustration of FilaLoom's working principle.

While the working principle is quite simple in concept, different applications of this mechanism each require their own approach to yarn path planning and sequencing of weaving and printing operations. These print strategies are discussed on a case-by-case basis in section 6.

3.3 GCode Generation

Yarn-embedded printing with FilaLoom requires custom GCode. To create compatible GCode, we developed a parametric design workflow in Grasshopper. Five different Grasshopper scripts were developed, each corresponding to a specific print strategy discussed in Section 6. While specifics of each script vary depending on the print strategy, all follow the same basic principles (Fig.8).

The script takes an input geometry and automatically derives a set of points from it. These points define the location of anchor pillars in the printed structure. The manner in which they are derived depends on the specific print strategy, with the user able to control parameters such as spacing and height. A curve is then generated along these anchor points, defining the path the yarn takes through the print. This toolpath curve is discretized into a series of closely-spaced points whose XYZ-coordinates are used to generate the GCode of the weaving sequences. Pick-up and parking sequences are then added to the weaving GCode; these are standard GCode lines that don't change depending on the weaving path. When using a double station, the script alternates between the different yarn depositors every n sequences, where n is user-defined.

The weaving GCode is then combined with the printing GCode. Some non-planar print strategies require custom GCode for the printing sequences as well, in which case both the printing and weaving GCode are generated in the same Grasshopper script and combined internally. These strategies are quite hands-free; the workflow for the user consists of inputting a base geometry, defining several parameters, and downloading the complete GCode file. The print strategies discussed in Section 6.1, 6.2, and 6.3 follow this workflow.

Other strategies use a conventional slicer to generate the printing GCode, which requires some additional steps from the user. After defining the anchor points, the script generates the weaving GCode, and modifies the original input geometry. The modified geometry includes anchor pillars at the anchor points and clearance to allow FilaLoom's needle to pass between these anchors. The user then exports and slices the modified geometry. During slicing, the user adds marker keys at layers where a weaving sequence should occur. These markers are used to identify the correct lines in the file, into which the weaving sequences are then manually inserted in a text editor. The print strategies in Section 6.4 and 6.5 use this conventional slicer workflow.

The resulting combined GCode allows a FilaLoom-enabled FDM-printer to automatically fabricate a yarn-embedded print. The STL files for FilaLoom's 3D-printable parts, assembly instructions, and Grasshopper scripts are available on GitHub.[1]

4 Design Space

By enabling the embedding of yarns, FilaLoom expands the FDM-printing design space by introducing new variables for the design of multi-material structures. These variables include an expanded material palette, novel fixation strategies, and opportunities in yarn positioning and finishing processes.

4.1 Materials

Compared to conventional FDM-printing, yarn-embedded printing gives an extra dimension when choosing materials (Fig. 9 left). In addition to selecting filaments for desired mechanical and aesthetic qualities, yarn selection introduces a range of additional variables. This includes the yarn's diameter; aesthetic qualities, such as texture, color, and finish; and mechanical properties like elasticity, strength and flexibility. These qualities can vary drastically between yarns, offering a wide range of options to design with. FilaLoom can embed up to two continuous yarns into a print, doubling these options.

The materials cannot be chosen in isolation however, as their combination determines the overall properties of the printed structure. These combined material interactions span adhesion, aesthetic- and mechanical properties. Adhesion between materials affects the structural integrity of the print. Aesthetic properties interact between materials to create contrasts and complementary effects. Mechanical properties such as the relative stiffness, elasticity, and strength of the yarn and filament interact to define how the print behaves under tension or compression. Designers must navigate this combined material space, balancing the aesthetic and mechanical properties of both yarn and filament to tune the properties of the print as a whole.

[1] `https://github.com/johaas-ams/filaloom`

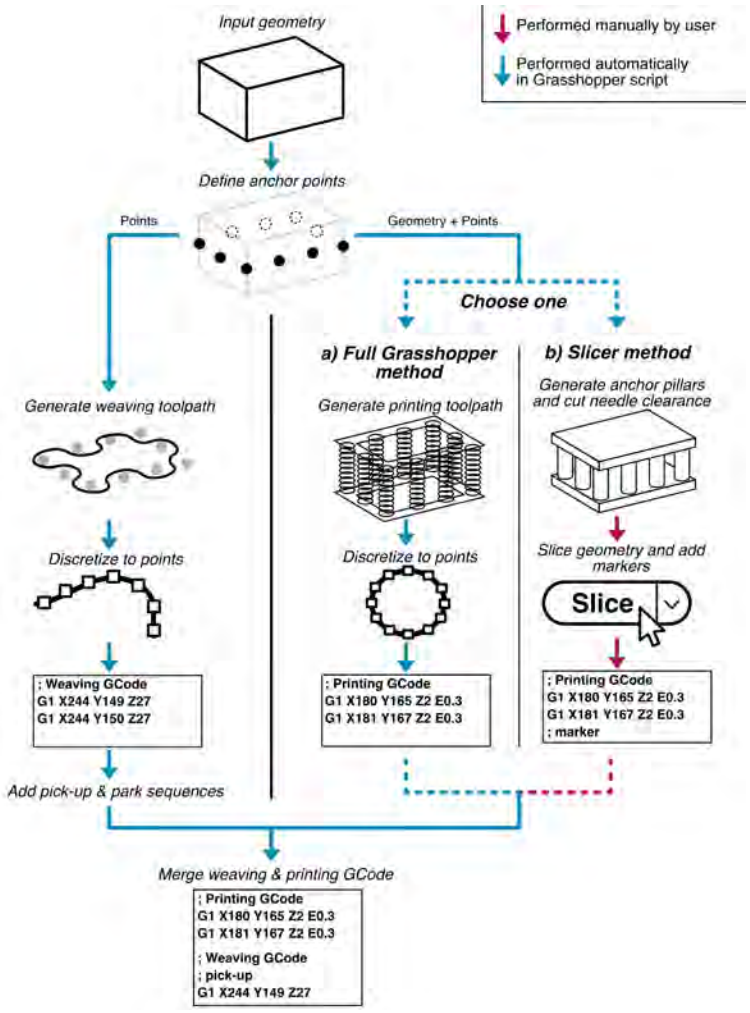


Figure 8: A flowchart of how FilaLoom-compatible GCode is generated, using either a fully integrated Grasshopper method, or a hybrid method using Grasshopper for the weaving GCode and a conventional slicer for the printing.

4.2 Fixation

When using FilaLoom, yarn can be fixed to FDM-printed parts in three different ways: friction, adhesion, and clamping (Fig. 9 right). Yarns are fixed through friction by wrapping the yarn around a pre-printed geometry with a hook, crevice, or channel. The geometry of the filament positions the yarn and controls the friction between yarn and filament to either allow it to glide in place similar to woven fabrics, or hold firmly in place. To increase yarn position stability, the printer can circle a joint several times to wrap the yarn around completely. Still, singular fixations are generally weak and easy to detach, but when combined together in a matrix they are able to create strong, tightly interwoven structures.

Yarns can also be attached to the print by extruding molten filament directly onto them; this 'glues' the filament and yarns together. The level of adhesion can vary widely depending on printing temperature, printing speed and material combinations [40, 41]. In general, yarns spun with staple fibers (e.g. cotton) demonstrate good yarn-filament adhesion due to their fibrous surfaces [37, 8].

Yarns can also be fixed by sandwiching them between two layers of filament, clamping them in place. Clamping requires the yarn to be positioned onto a printed surface, after which filament is extruded over it. If yarn-filament adhesion is high, this simultaneously fixes the yarn through adhesion, resulting in a strong bond. Otherwise, a clamped yarn can slide and rotate about the printed part.

Note that in order to fix yarns into a print through adhesion or clamping using FilaLoom, they should first be held in position using anchors (a type of friction-based fixation), after which the depositor is parked, and the nozzle can print over the yarn.

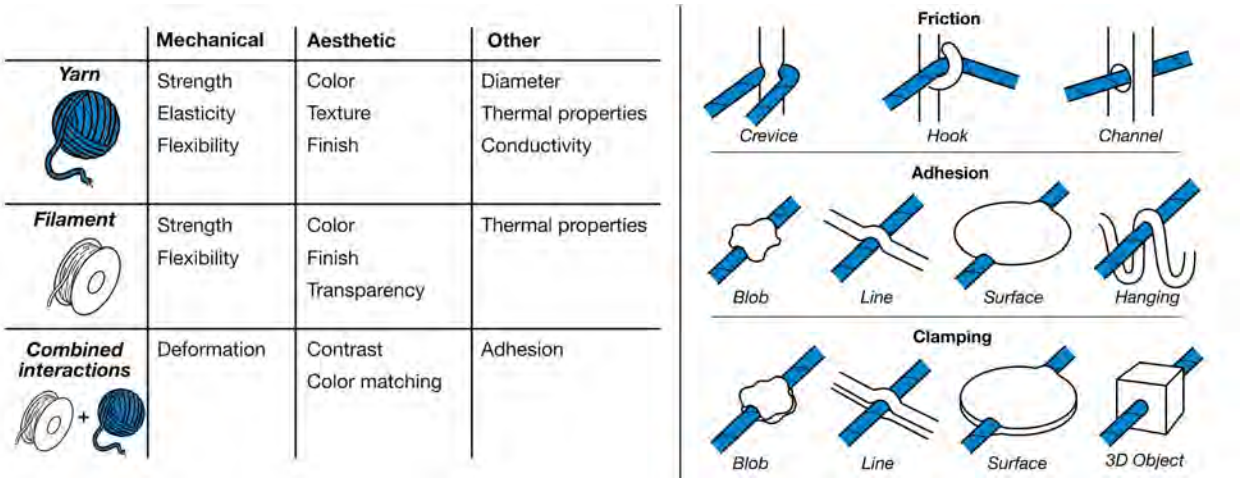


Figure 9: **left**: Material design variables for yarn-embedded FDM-printing. **right**: Yarn fixation types for yarn-embedded FDM-printing.

4.3 Yarn Position

To position yarns in a print, the yarn is strung between two anchor points. Yarn positioning in FDM-printing can be either in-plane or out-of-plane relative to the print volume [8]. FilaLoom allows yarns to be placed in both orientations. Due to the layer-by-layer nature of FDM-printers, positioning a yarn in-plane is straightforward: the yarn can be embedded by simply adding a pick-up $\rightarrow$ weave $\rightarrow$ park sequence in between layers of conventionally sliced GCode, leaving the conventional printing order uninterrupted. For out-of-plane positioning, FilaLoom's yarn depositor needle dips 25 mm below the nozzle. This allows yarns spanning less than 25 mm vertically to be spun in a single weaving sequence, given there are no roofs or walls blocking the needle's travel path. For larger vertical distances, the yarn is first anchored at the lower point, the depositor parked, subsequent layers printed, and the yarn then stretched to the higher point. In this last case, weaving can only progress from low to high, as the print head blocks the needle from reaching parts that lie more than 25 mm below the highest printed layer.

When embedding a network of yarns into the print, such as to fabricate textile-like structures, each span of yarn also has a position relative to the others. This *yarn layout* can be described by the yarns' relative angles to each other as well as the spacing between them. It influences the structure's strength, deformation behavior and tactile qualities.

4.4 Post-processing

After the print is finished, the user can undertake post-processing steps. FilaLoom can only embed continuous yarns; these yarns can be cut after printing to change their length or disconnect separate parts. The cut ends can then be unraveled to create a frayed, hair-like texture, or melted to seal off the ends (for thermoplastic yarns). The print can also undergo post-print deformation by applying tension to the yarns, creating interactive structures. Aesthetic qualities such as texture, color or finish can be adjusted by sanding, painting or embroidering the prints. Lastly, separate parts can be combined into a larger structure by joining, gluing, or sewing them together.

5 Guidelines

The following guidelines provide recommendations on the use of FilaLoom, covering suitable yarn types, operating parameters, and known constraints of the system.

Suitable Yarns. FilaLoom is suitable for a large variety of yarn, from thin lock thread to bulky knitting yarn. The dispensing needles used to place the yarns are standard parts which come in different diameters, making it possible to switch needles to suit the yarn diameter. This project used 50 mm-long 15G and 11G needles with inner diameter of 1.34 and 2.5 mm respectively, switching between them to suit the diameter of the yarn used. Most samples were fabricated with 1.5 mm-diameter 475 tex cotton yarn using a 15G needle, as well as 0.6 mm-diameter 100 tex polyester yarn using an 11G needle. Because the mechanism relies on yarns being easily pulled along in any direction, stiff yarns such as metal wire or straw are not suitable.

Tension. Tension in the yarn can be adjusted by screwing the bolt on the axle. This can be used to tune the tightness of the yarn within the print. Low tension leaves more slack in the yarn, higher tension pulls the yarn tighter against the anchors, increasing structural integrity and path accuracy. However, excessive yarn tension risks deforming the print, breaking it, or ripping it from the print bed. The suitable tension level for a specific print depends on multiple parameters, including yarn type, yarn layout, print geometry, and filament material. It is therefore difficult to give an exact guideline for setting the appropriate tension, and users must develop practical intuition through repeated use. The system is not very sensitive to tension however, with most applications, including those discussed in section 6, printing well across a range of tension settings. FilaLoom cannot apply tension to the yarn dynamically, so tension remains constant throughout the printed structure, unless manually adjusted.

Speed. FilaLoom has a maximum operating speed of 25 mm/s when embedding yarns; beyond this speed, the accuracy of yarn placement is affected. The pick-up and parking sequence can be performed consistently at speeds up to 133 mm/s; operating faster affects the consistency and risks damage to the printer. At this speed, a pick-up sequence takes approximately 3 seconds, and a parking sequence approximately 12 seconds, which includes a nozzle wipe and priming strip before continuing printing. The station also adds mass to the print bed, increasing the momentum of the Y-movement in bed-sling printers; extremely fast Y-moves should therefore be avoided. Standard operating speeds for PLA and TPU are well within safe range.

Precision. The yarn depositor needle needs 1.5 mm horizontal clearance to move smoothly through a print. For example, the 15G needle used in testing has an outer diameter of 1.84 mm, meaning the route it takes through a print should be at least 3.34 mm wide. For consistent placement, the height of an anchor needs to be larger than the yarn diameter. Anchors also require a horizontal crease, so the yarn stays in the desired position after it is placed. This crease should be approximately half the yarn diameter.

Reach. The yarn depositor needle dips 25 mm below the nozzle of the printer; thus yarns can be placed 25 mm deep into a print, given the roof of the structure is open. A longer needle could potentially be used to increase this reach, but this comes at the expense of the needle being easier to bend during use, which affects the accuracy of yarn placement.

Build dimensions. The Bambu Lab A1 has a print volume of 256 $\times$ 256 $\times$ 256 mm. The station covers an area on the print bed of 30 $\times$ 106 mm, on which printing is not possible. There is also a clearance zone directly behind the station where nothing can be printed because the X-axis would bump into it during pick-up- and parking maneuvers. Because the yarn depositor's needle is positioned at a 40 mm offset from the nozzle in the Y direction and 25 mm in the Z direction, the yarn cannot reach Y-positions further than 222 mm, or Z positions further than 231 mm. Figure 10 shows the dimensional restrictions when using FilaLoom. The usable build volume within which both printing and weaving are possible is 54% of the printer's total build volume.

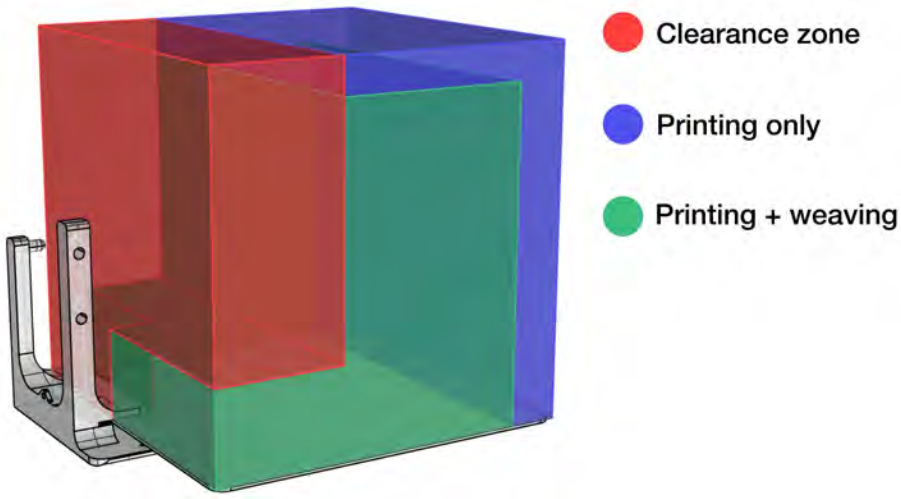


Figure 10: Usable print volume: Build volume restrictions when using FilaLoom on the Bambu Lab A1.

Bed adhesion. Weaving yarn into a print exerts a horizontal force on the printed structure. If adhesion between the print and build plate is insufficient, this force can cause the print to detach from the build plate. Enabling brims on printed parts is therefore recommended, especially when printing small parts or when yarn tension is set too high.

Slack. Each time the depositor is parked after a weaving sequence, the yarn is stretched from its last anchor point in the print to the station at the edge of the build plate. When the depositor is brought back towards the print in the next weaving sequence, the yarn that was previously stretched between the station and print will be longer than the distance it needs to span, causing it to go slack. This slack can make accurate yarn placement at the start of this next weaving sequence difficult. To mitigate this issue, the yarn can be wound around the hook on the station before parking, keeping it tensed at the start of the next sequence. This ensures consistent positioning, but creates a loop between station and print that must be cut in post-processing. It also prevents the yarn from running continuously across multiple weaving levels. Alternatively, increasing tension can cause the yarn to self-correct, sliding into the right position before the spool unwinds further. This requires careful tuning of yarn tension and is less consistent than winding it around the station, but is necessary for out-of-plane yarn positioning.

6 Print strategies

This section showcases the application potential of FilaLoom by presenting several custom print strategies in which it enables the fabrication of FDM-printed parts with yarns embedded to create novel structures, textures, and interactions. The Grasshopper scripts used to generate GCode for fabricating these structures can be found in the GitHub repository.

6.1 Vertical woven objects

Vertical woven objects (Fig. 11) are structures consisting of vertical pillars with yarns woven between them horizontally. The pillars contain crevices for the yarn to sit in, holding it in position in a woven network. Because the structure is fabricated in the Z-direction these woven objects can take various three-dimensional shapes.



Figure 11: A collection of vertical woven objects fabricated with FilaLoom.

Our printing strategy (Fig. 12) begins with printing a flat brim, providing a stable base on which the pillars can stand. The pillars are divided into segments that are printed in a single helix-motion over multiple layers, a segment being the section of a pillar featuring one crevice. For each pillar, the nozzle prints one segment, then retracts sideways and moves down to print the corresponding segment of the next pillar. Compared to a layer-by-layer approach, printing in

segments minimizes oozing between adjacent pillars, which could otherwise block the needle from passing between them. Once a segment has been printed for every pillar, the depositor is picked up, yarn is woven between the segments, and the depositor is parked. The printer then proceeds to print new segments on top of the previous, where the same sequence repeats, gradually building up the woven structure. Once the full structure is printed, a brim is printed across the tops of the pillars, closing the structure.

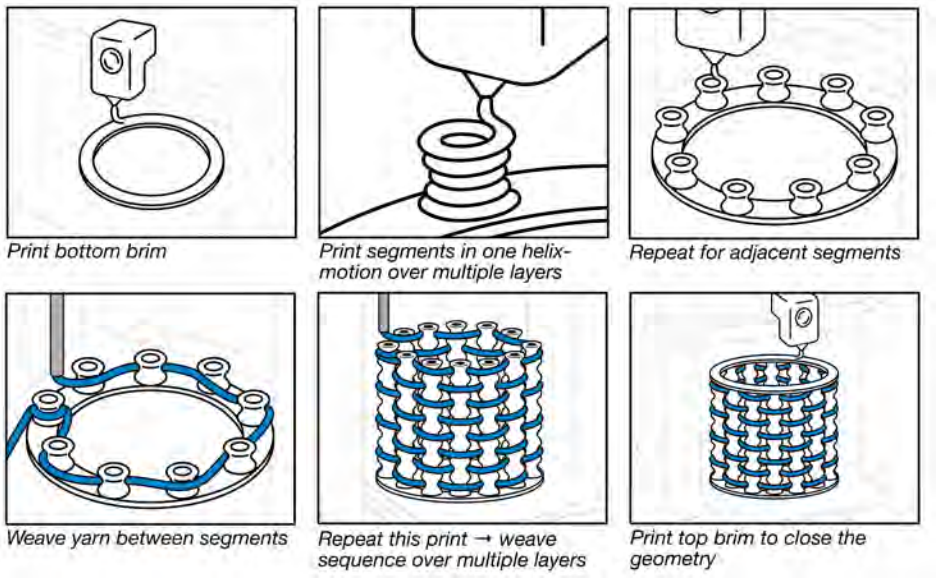


Figure 12: Print strategy for fabricating vertical woven objects.

The Grasshopper script which generates GCode for printing vertical woven objects takes a lofted surface as input geometry. Based on the shape of this surface, it generates the toolpath for printing the pillar structure, as well as the weaving toolpaths which snake between the pillars. These toolpaths are then converted to GCode, merged, and sequenced in the correct order, outputting a ready-to-print GCode.

Besides geometry, various other parameters can be tuned (Fig. 13). To change the weave density, one can define the radii of the pillars, horizontal distance between pillars (warp density) and vertical distance between yarns (weft density). The weave pattern is defined by a binary sequence, enabling weave structures such as plain weave, twills, and satin.

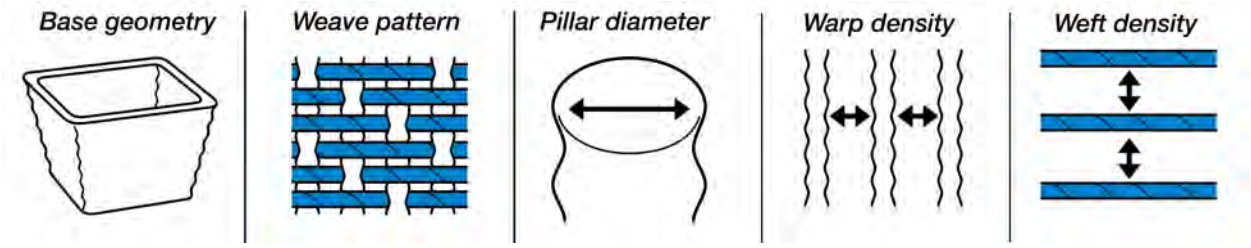


Figure 13: Design variables of vertical woven objects.

6.1.1 Examples

The following examples demonstrate a range of vertical woven objects fabricated with FilaLoom, each highlighting a different capability. All samples were fabricated using 1.5 mm diameter cotton knitting yarn.

Weave patterns. FilaLoom's binary weave input allows different weave patterns to be produced. Figure 14a shows a set of four hexagonal samples featuring plain weave, 2x1 twill, 3x1 twill, and satin, all fabricated with identical parameters, isolating the effect of the weave pattern on the surface texture of the object.

Multi-color weaving. Because FilaLoom can embed up to two yarns, alternating between them across weft layers produces multi-color woven structures. Figure 14b shows a bowl-shaped object woven with blue and lilac yarn, alternating at every weft layer.

Mid-print brims. When pillar diameter is small, pillars can become unstable with increasing height, particularly in flexible materials such as TPU, where the pillars deform more easily under the tension of the yarn. Inserting an intermediate brim mid-print connects the pillars laterally, stabilizing them and allowing the structure to be built up further. The exact height at which a brim becomes necessary depends on pillar diameter, material, yarn tension, and weave density, so no fixed rule can be given. Figure 14c shows a double-layered TPU95A print demonstrating this strategy.

Non-developable surfaces . FilaLoom's ability to weave in three-dimensions enables fabrication of 3D geometries that are not possible with traditional weaving. Figure 14d shows a lampshade with a curved non-developable surface, demonstrating the geometric freedom of vertical woven objects, as well as their application potential in decorative objects.

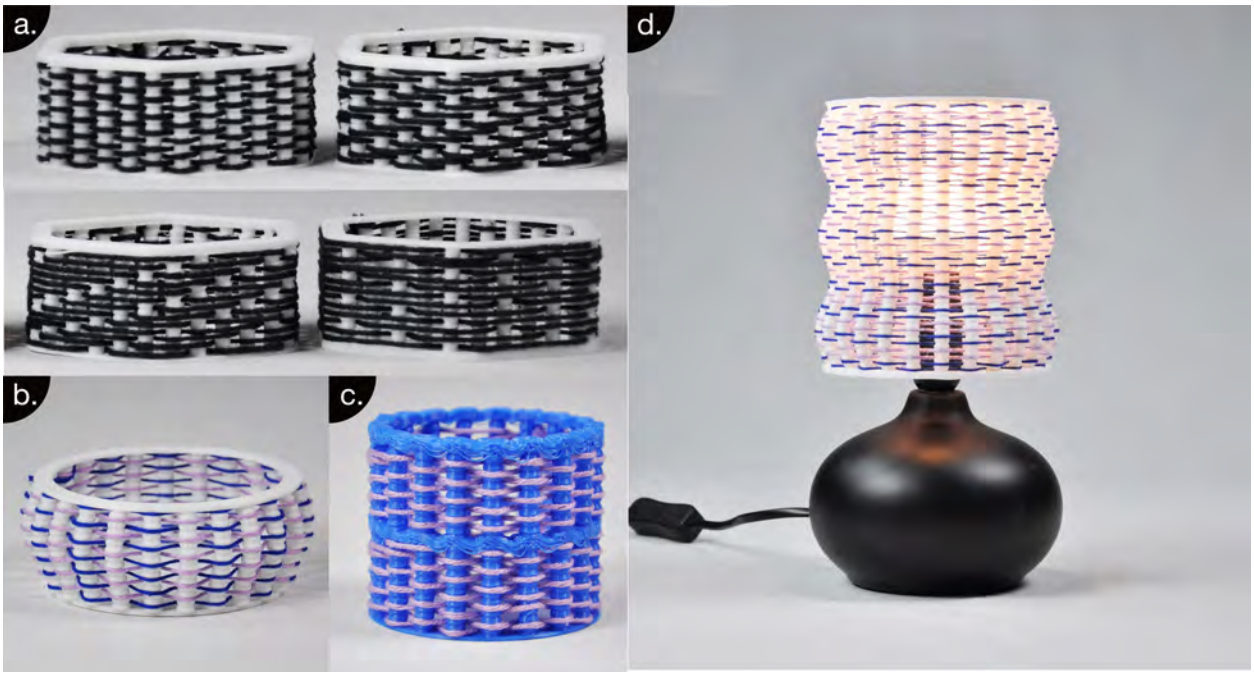


Figure 14: Vertical woven object examples: (a) four hexagonal prints with plain weave, 2x1 twill, 3x1 twill, and satin weave patterns; (b) a woven print with alternating blue and lilac yarn; (c) a print with a mid-print brim; (d) a lampshade with a non-developable curved surface.

6.2 Planar semi-woven textiles

Planar semi-woven textiles are flat, flexible hybrid textiles consisting of two printed filament layers with a network of yarn sandwiched between them (Fig. 15). The filament layers form a semi-warp, while the embedded yarn forms a semi-weft. The textile can be printed directly to its final shape, and feature wefts of various shapes and orientations, while exhibiting tactile and aesthetic qualities of yarn-based textiles.

The print strategy for fabricating semi-woven textiles (Fig. 16) begins with a bottom layer consisting of the semi-warp and a brim printed in the outline of the desired shape. Short anchor pillars are then printed on top of the brim, after which the yarn is woven between these anchor points in a zig-zag path from one side of the shape to the other, resulting in parallel sections of yarn which form the semi-weft. A top warp layer is then printed directly over the yarn, clamping it in place, and a top brim layer is printed over the anchors to conceal them.

A Grasshopper script was developed to generate custom GCode for this fabrication strategy. The script takes two curve inputs: a closed planar curve defining the outline of the textile, and a network of planar curves defining the warp layout,



Figure 15: A collection of planar semi-woven textiles fabricated with FilaLoom.

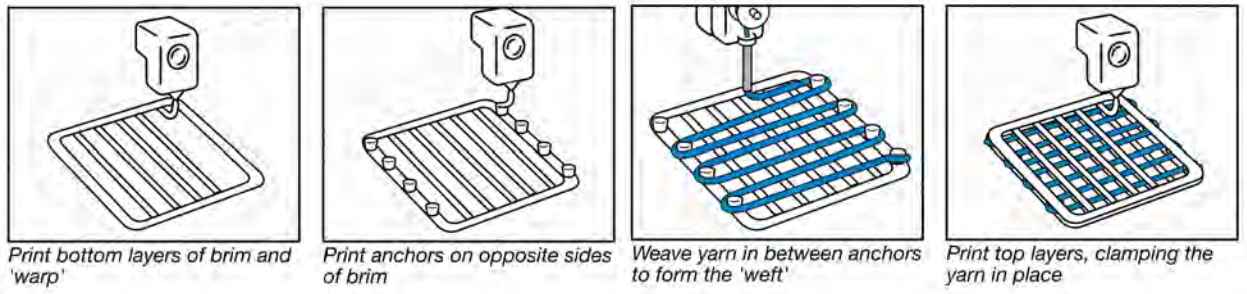


Figure 16: Print strategy for fabricating semi-woven textiles.

as well as a number input defining the spacing between adjacent yarns. From these, the script generates the anchor positions, weaving toolpaths and printing toolpaths automatically, exporting a complete GCode file.

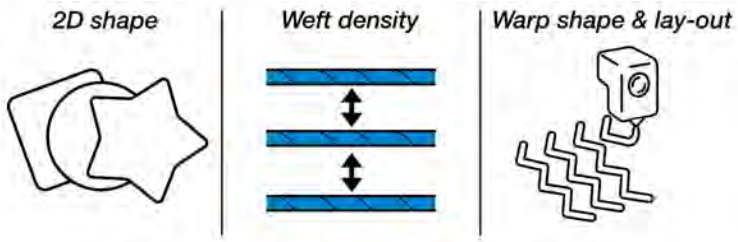


Figure 17: Design variables for planar semi-woven textiles.

6.2.1 Examples

The following examples demonstrate a range of planar semi-woven textiles fabricated with FilaLoom. All samples were fabricated from TPU90A filament and 1.5 mm diameter cotton knitting yarn.

Shapes. Because the yarn is spun from anchor points within the textile's outline, semi-woven textiles can be fabricated in any 2D shape. Figure 18a shows three samples, a square, a circle and a freeform blob, demonstrating the potential of this approach for producing textiles in custom shapes without cutting or waste.

Warp patterns. As opposed to traditional weaving, printed warps need not be straight or perpendicular to the yarn direction, allowing great expressive freedom. Figure 18b shows three samples, each featuring a different warp layout: a curved pattern, a zigzag pattern, and a pattern spelling out text. Such non-straight warp geometries can also be utilized to create dynamic deformations under tension [37].

Woven baskets. Different fabrication strategies can be combined within a single print to create more complex objects. Figure 18c shows a basket combining a planar semi-woven bottom with a vertical woven body, demonstrating the potential of FilaLoom to create functional soft objects.

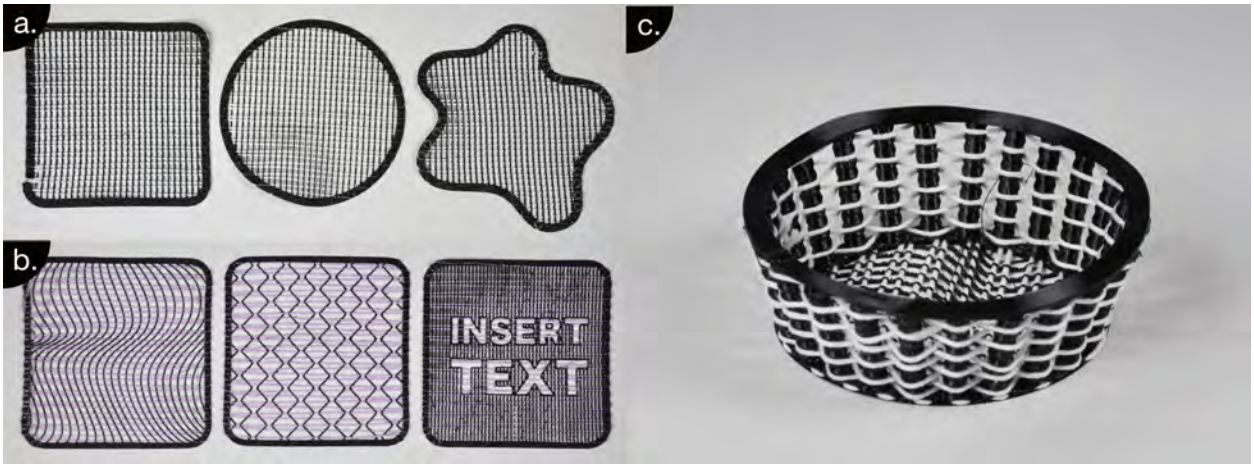


Figure 18: Planar semi-woven textile examples: (a) three textiles in square, circle, and freeform shapes; (b) three textiles showing curved, zigzag, and text-based warp patterns; (c) a 3D woven TPU basket combining a planar semi-woven bottom with a vertical woven body.

6.3 Loop-based spacer fabrics

Loop-based spacer fabrics are voluminous hybrid textiles fabricated by extruding filament over a web of spun yarns at an elevated height. As the extruded filament droops into the gaps between the yarns, it forms loops in the Z-direction, producing a textured surface with loops extending from the fabric in a three-dimensional pile (Fig. 19).

To fabricate this structure, yarns are woven at an elevated height through a pre-printed tool placed onto the print bed (Fig. 20). Printing this tool separately and reusing it across multiple prints mitigates waste, but it could alternatively be printed as a sacrificial part immediately before weaving. The depositor is then picked up and yarns are woven through the tool from side to side, forming a semi-weft. After, the depositor is parked and lines are printed perpendicular to the yarns. Gravity causes the filament to fall into the gaps between adjacent yarns, forming loops (Fig. 21). For this method to succeed, the lines should be printed at the exact same height as the yarns; when this is not the case the filament will coil up [42] instead of forming loops.

A Grasshopper script was developed to generate GCode for fabricating these spacer fabrics. The script uses a fixed linear layout for the weaving and printing toolpath and converts these to GCode, allowing extrusion to be tuned. For this fabrication method, extrusion is the primary variable determining the character of the resulting fabric, as it directly controls loop height. To quantify this relationship, a series of tests were performed using a 5 mm yarn gap and a 0.4 mm nozzle, revealing a linear relationship between extrusion and loop height for both PLA and TPU:

$$\text{PLA: Loop Height} = 24 \times \text{Extrusion} + 8 \tag{1}$$

$$\text{TPU: Loop Height} = 20 \times \text{Extrusion} + 7 \tag{2}$$

Here, Loop Height is given in millimeters and Extrusion in millimeters of filament fed through the extruder per millimeter of travel distance. These formulas allow to predict and control the loop height (Fig. 22). For optimal loop homogeneity, printing at 230°C and 2 mm/s is recommended.



Figure 19: A loop-based spacer fabric fabricated with FilaLoom.



Figure 20: A weaver tool used to hold yarn at an elevated height for fabricating loop-based spacer fabrics.

6.3.1 Examples

Several variations of the technique were developed, showcasing different ways of applying this print strategy. All examples were fabricated with TPU95A and 1.5 mm diameter cotton knitting yarn.

Homogeneous . Homogeneous textures are the 'standard' way of applying this technique. The filament is extruded at a consistent rate, creating evenly sized loops which form a carpet-like texture (Fig. 23a).

Molded. At high extrusion rates, loops can extend far enough down to reach the print bed (Fig. 23b). This causes the ends of the loops to coil up and intertwine with each other, creating a dense surface texture connected to the yarns by thin filament strings. This effect can be manipulated by placing a mold on the print bed, as the filament will conform to the shape it falls onto. The structure behaves similarly to conventional spacer fabric, softly deforming when pushed in.

3D images. By varying the extrusion throughout a print, localized differences in loop height can be used to create three-dimensional images. We mapped the brightness values of a grayscale image to extrusion values to translate the image into a surface texture. A fabric featuring an image of a smiley was fabricated (Fig. 23c), demonstrating sufficient contrast in loop heights to create visible images. In the 5x5 mm grid size used, images can be created with a quality of approximately 25 DPI.

6.4 Soft textures

Rather than using yarn as part of the material structure, FilaLoom can also embed yarn into the surface of regularly printed objects to create soft yarn textures (Fig. 24). By modifying the object's geometry to include anchors and

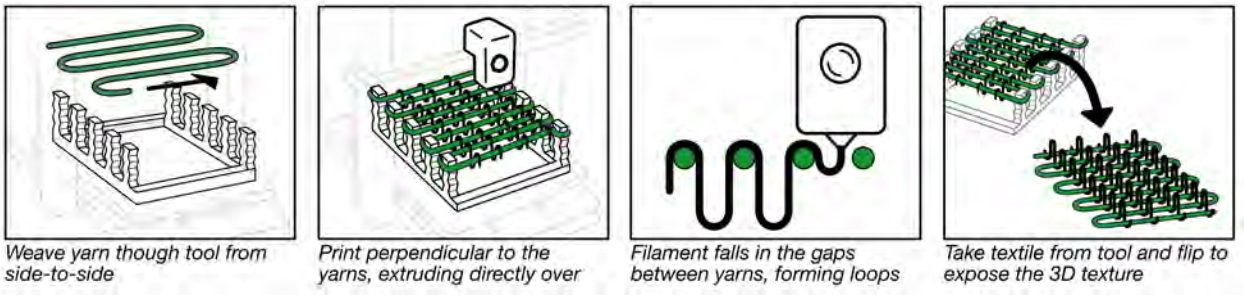


Figure 21: Print strategy for fabricating loop-based spacer fabrics.

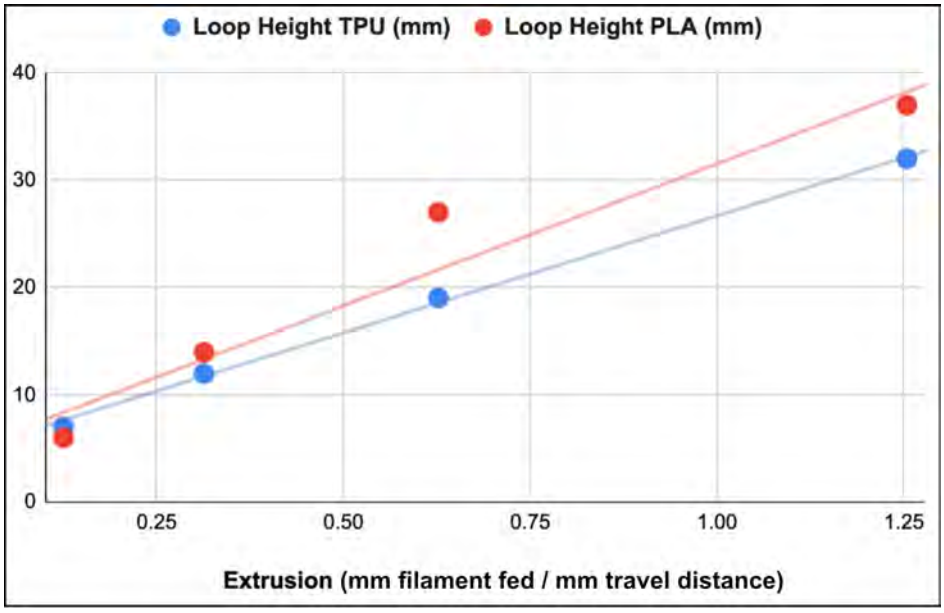


Figure 22: The relation between extrusion and loop height when fabricating loop-based spacer fabrics.

channels, yarn can be woven into the object during printing, adding tactile and aesthetic qualities to conventional FDM-prints.

To embed a yarn texture, an object is modified in two ways: channels are cut into the object to allow the depositor needle to pass through, and anchor structures are added to the surface where texture is to be applied, the geometry of which depends on the specific texture being added. These different texture types are described in section 6.4.1. The print strategy for embedding yarn textures is shown in Figure 25. The structure is printed up to the first weaving level, after which the depositor is picked up and yarn is woven through the channel and anchor structures. The depositor is then parked, and a blob is printed into the channel, fixing the yarn in place. Printing then continues, completing the structure up to the next weaving level. After which, the depositor is picked up again, the yarn is woven through the print, and guided back out through a second channel. The depositor is then parked, and this channel is sealed in the same way.

A Grasshopper script was developed to generate GCode for the weaving sequences and assist with geometry modifications. The printing GCode is generated by slicing the geometry in a conventional slicer, after which the weaving sequences are manually inserted as described in section 3.3. Several parameters can be varied when applying this method (Fig. 26). The placement and dimensions of the textured area, the vertical distance between yarns which determines the density of the texture, and the texture type, which determines the overall appearance of the surface. These different texture types are described in the examples below.

6.4.1 Examples

Three methods for creating surface textures were developed: woven, wrapped, and loop-pile. All samples were fabricated from PLA and 1.5 mm diameter cotton knitting yarn.

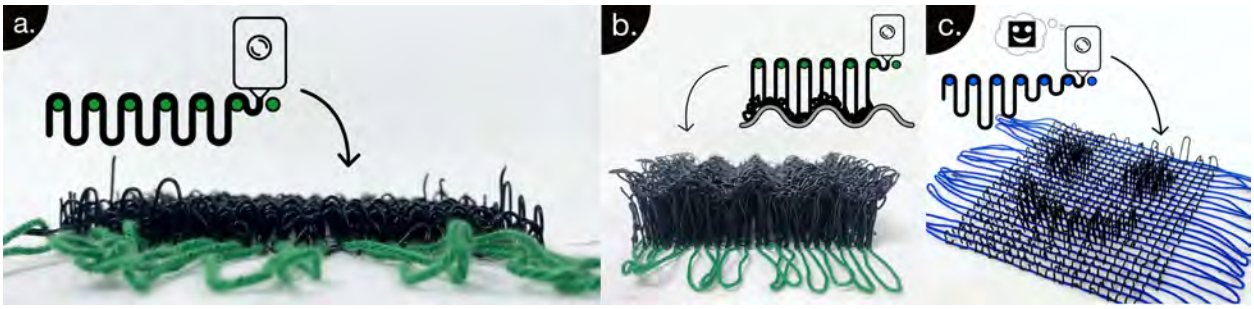


Figure 23: Loop-based spacer fabric examples: (a) homogeneous loops forming a carpet-like texture; (b) molded loops conforming to a wave-shaped mold; (c) a 3D image of a smiley face created through localized variation in loop height.

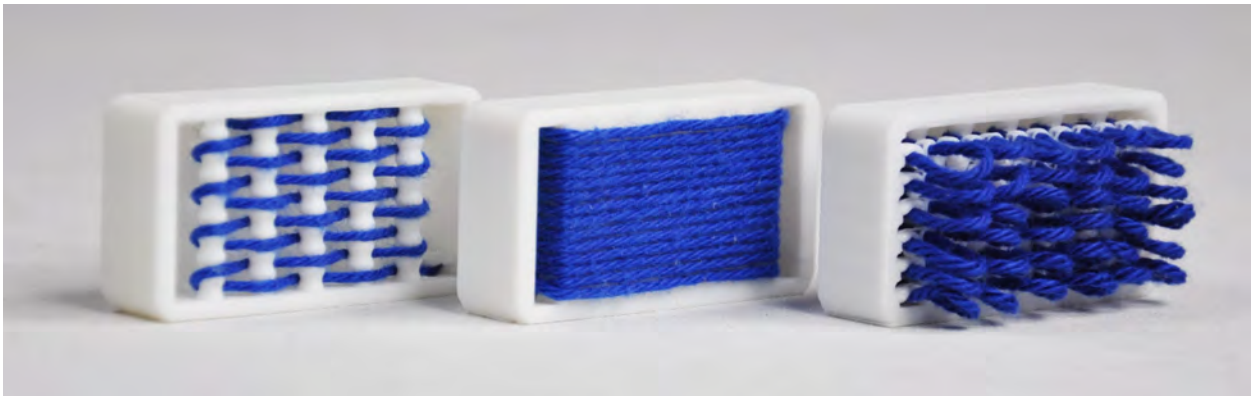


Figure 24: Printed brushes with different soft yarn textures

Wrapped. Smooth yarn textures are created by wrapping the yarn around one of the walls in the print (Fig. 27a). The walls have clearance on either side for the needle to pass through; the needle circles around this wall many times, wrapping it in yarn to create a soft uniform texture.

Woven. To fabricate woven textures, the surface of a geometry is replaced by vertical pillars, through which the yarn is woven to create a textile-like surface texture (Fig. 27b). This is the same general strategy as is used to fabricate vertical woven objects, but embedded into the surface of another geometry.

Loop-pile. Loop-pile textures (Fig. 27c) use a second, sacrificial row of pillars opposite the surface to act as a support structure for the yarn. The yarn is woven in a zigzag pattern, alternating between a pillar inside the surface and one on the opposite, sacrificial row. After every weaving layer, the loops are fixed into the surface by printing a line over them, clamping the yarn in place. This method creates a raised loop-pile surface. The pile height can be adjusted by changing the distance between the two rows of pillars. Once the sacrificial pillars are removed, the loops can also be cut to create a cut-pile, hair-like texture. Unlike wrapped and woven textures, loop-pile textures cannot be fabricated across multiple layers in a single sequence because each layer of yarns must be clamped in place by printing a line over them.

6.5 Cable-driven mechanisms

To increase interactivity within prints, FilaLoom can be used to fabricate cable-driven mechanisms (Fig. 28). To fabricate such mechanisms, a yarn is routed through a geometry containing motion primitives and is fixed at two ends, one inside the geometry, and one in a detached part that acts as a puller. Pulling this detached part transmits a force through the yarn, actuating the motion. FilaLoom enables the integration of such mechanisms into textile-like objects, adding an interactive dimension to these prints.

To run the yarn through a cable-driven mechanism, a channel is cut through the motion primitives. At the ends where the yarn should be fixed, a short cylinder-shaped anchor is placed inside the object. Around this anchor, a clearance is cut out of the geometry to allow the yarn depositor's needle to reach the anchor. The process of printing cable-driven mechanisms is visualized in Figure 29. The object is printed as normal up to the height of the channel and anchors.

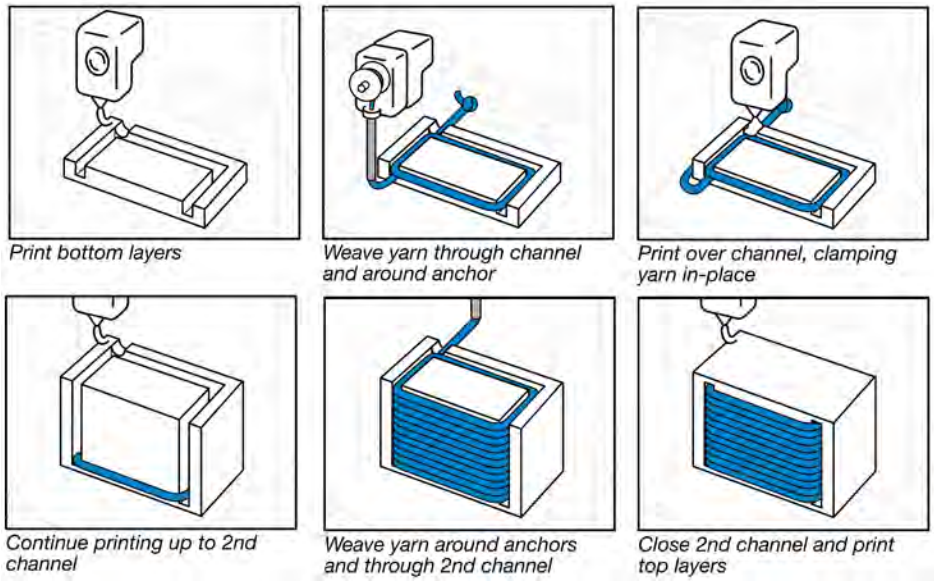


Figure 25: Print strategy for embedding yarn textures into a print

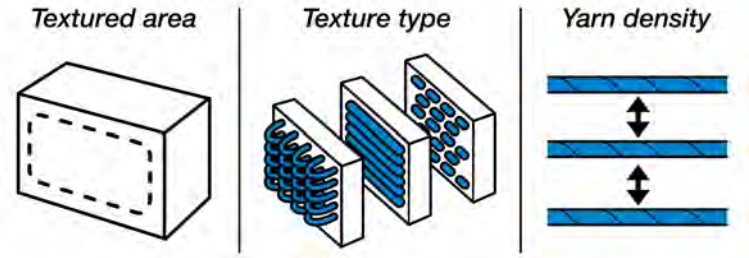


Figure 26: Design variables for soft yarn textures.

Before any roofs are printed over them, a weaving sequence starts; the depositor is picked up, the yarn is wrapped around the first anchor, guided through the channel in the motion primitives, and wrapped around the second anchor. This routes the yarn from one end of the cable-driven mechanism to the other. The depositor is then parked, after which the hollow areas surrounding the anchors are filled in by printing over them, clamping the yarn tightly. This 'wrapping plus clamping' technique creates a strong connection able to withstand the pulling force required to actuate the mechanism.

To fabricate cable-driven mechanisms with FilaLoom, one first models a cable-driven geometry, and draws a curve running through this geometry from end-to-end. These are then inserted into a Grasshopper script. The curve defines the toolpath the yarn takes through the motion primitives, and is used to generate the weaving GCode. The script also modifies the geometry by cutting a channel along the curve, and places anchors at the ends of the curve. The geometry is then exported and sliced, after which the weaving sequence is manually inserted into the GCode as described in section 3.3. The primary design variables for cable-driven mechanisms are the motion primitives and overall geometry of the printed object. As this is specific to the object being designed rather than the yarn-embedding process, it falls outside the scope of this work. FilaLoom allows yarns to be routed through a print in any horizontal path, potentially enabling fabrication of various cable-driven mechanisms.

6.5.1 Examples

Figure 30 shows two examples of cable-driven mechanisms embedded into a planar semi-woven textile, resulting in textile sheets which exhibit bending and coiling motions respectively. The structures were fabricated in a single uninterrupted print, demonstrating FilaLoom's potential for the automatic fabrication of textiles with interactive properties.

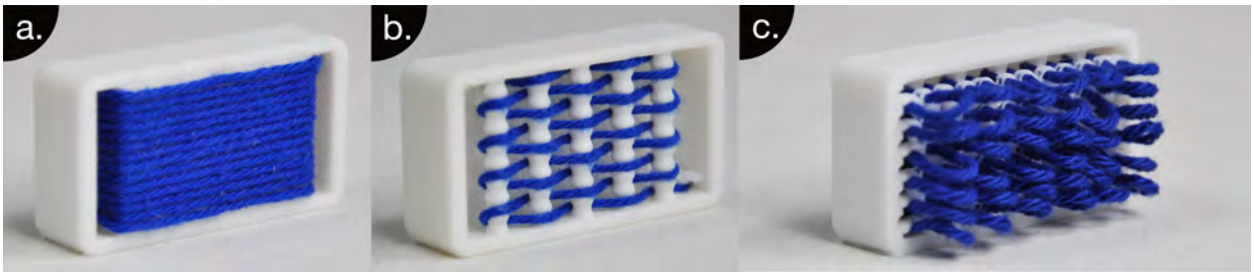


Figure 27: Soft yarn texture examples: (a) smooth texture created by wrapping yarn around an outer wall; (b) woven texture created by weaving yarn through pillars; (c) loop-pile texture created by weaving yarn in between surface and sacrificial pillars.

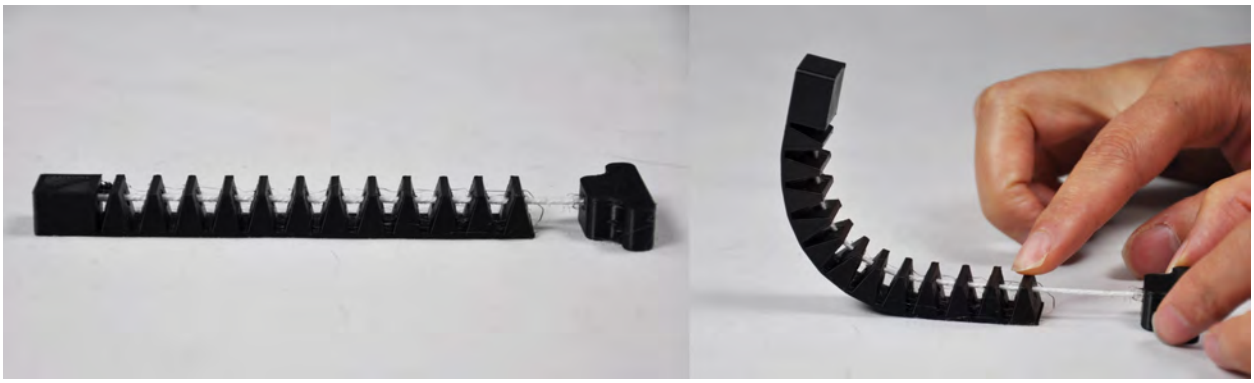


Figure 28: A cable-driven mechanism fabricated with FilaLoom, with bending motion actuated by pulling the puller.

7 Discussion, Limitations & Future Work

FilaLoom demonstrates that automatic yarn-embedding in FDM-printing is possible with only simple printer modifications. The fabrication strategies presented expand the possibilities of FDM-printing, enabling fabrication of objects with increased textileness and interactivity. We next discuss key strengths and limitations of our work and identify directions for future research.

7.1 Accessibility

Making digital fabrication tools accessible to a broader audience is a recurring motivation in HCI [43, 44, 45]. FilaLoom is a simple mechanism aimed at designers, researchers, and artists working with digital fabrication in fields such as textiles, product design, and fiber arts. The mechanism runs entirely on the printer's own movement, requiring no additional electronics. It is mostly 3D printable, and its three non-printable parts are widely available at online retail stores. Assembly makes no permanent changes to the printer; the entire set-up can be attached and detached in a few minutes. These characteristics give FilaLoom a low barrier to entry for experimenting with yarn-embedded printing. Adapting FilaLoom to other FDM-printers would require modifying the dimensions of both the print head cover and station, which are currently fitted to the Bambu Lab A1. This is relatively simple for anyone with CAD skills. The working principle and the developed printing strategies remain functional across conventional FDM-printers, both bed-sling and core-XY models. For core-XY models, the station could be placed on the frame of the printer to free up the entire print volume.

While FilaLoom itself is accessible and straightforward to implement, creating compatible GCode is currently achieved through developing and using custom Grasshopper scripts, requiring Grasshopper expertise and a general understanding of GCode. While the scripts for the print strategies described in sections 6.1, 6.2, and 6.3 produce complete GCode files, the workflows for those in section 6.4 and 6.5 require manually pasting custom GCode sequences in a sliced GCode file, which is time-consuming and sensitive to mistakes. Furthermore, there is currently no general-purpose tool to help develop GCode for new yarn-embedded print strategies. Future work could address this through a dedicated software that allows one to construct yarn toolpaths and sequence the printing- and weaving order through a user-friendly interface, enabling accessible generation of GCode for various yarn-embedded print strategies.

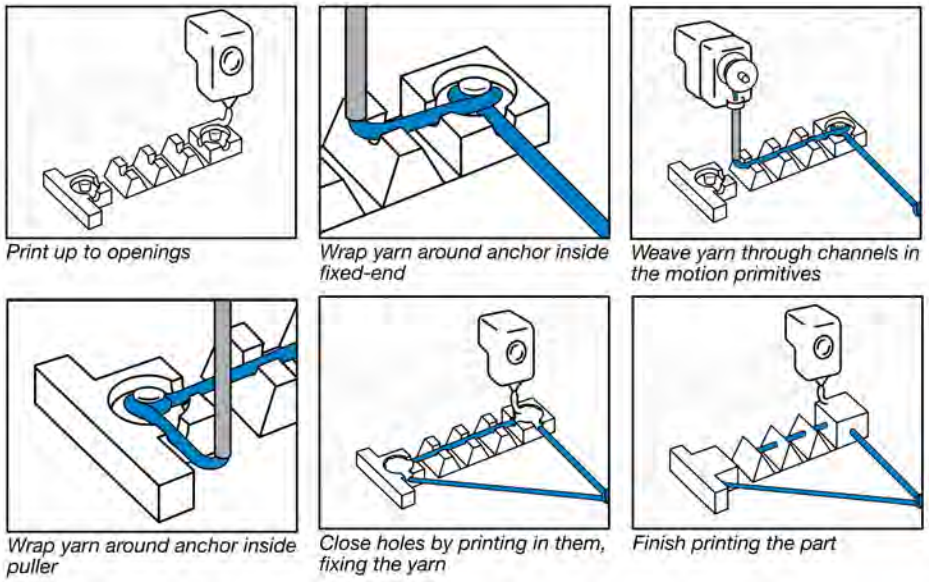


Figure 29: Print strategy for fabricating cable-driven mechanisms.

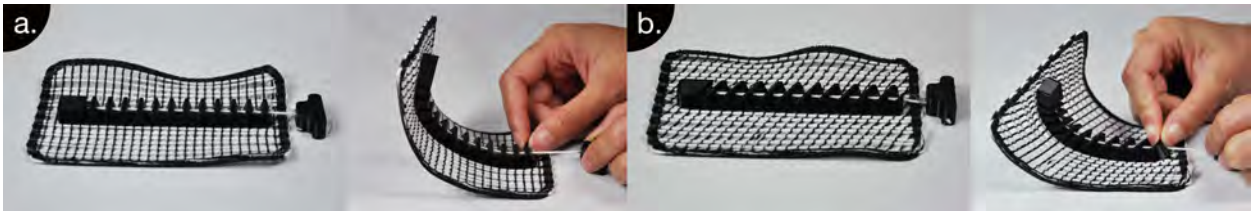


Figure 30: Two textiles with cable-driven actuation: (a) a bending textile at rest and actuated; (b) a coiling textile at rest and actuated.

7.2 Mechanism Constraints

FilaLoom has several practical constraints worth discussing. FilaLoom's tension control is constrained by the tension inherent to the system. This inherent tension is caused by friction between the yarn and the depositing needle, both through contact with the inner channel and the sharp turn the yarn takes when fed through the needle tip. In practice, this friction is sufficient to sustain yarn tension for most applications. However, it also makes the system incompatible with off-the-shelf tension control components, as the additional tension introduced would compound and result in excessive tension that causes the print to break or deform. The current tension control, a spring that pushes a washer against the yarn spool, was therefore designed to have minimal baseline tension. One limitation of this DIY component is that yarn tension can vary slightly within a print. This has caused only few issues in our demonstrated applications, but could restrict FilaLoom's use in tension-sensitive applications such as prints with elastic yarn. Future work could focus on reducing friction within the system to make it compatible with more precise tension control components, as well as study the implication of tension as a design variable more in-depth through dedicated parameter sweeps.

As described in the guidelines, each parking sequence creates slack in the yarn. Ideally, this slack would be taken back in, eliminating the need for workarounds. Several mechanisms were prototyped during development, but each would increase yarn tension to levels high enough to break the print or detach it from the build plate. Future work could look into mechanisms that take up slack without increasing yarn tension.

Another constraint is the number of yarns FilaLoom can embed. The system currently supports embedding two different yarns within one print, enabling switching between different colors and properties. However, many applications could benefit from more yarns within one print. Extending the usable number of yarns would simply require fitting additional stations onto the build plate. Future work could optimize FilaLoom's dimensions to accommodate more stations within the available build volume.

Furthermore, as FilaLoom cannot cut the yarn after embedding it into the print, it is only able to embed continuous yarns. Using short yarn segments is therefore only possible by cutting the yarn in post-processing. Future research could look into techniques for cutting yarns on the printer, such as utilizing the nozzle to burn through the yarns [37].

Finally, as FilaLoom's tool-changer is a printable DIY system, it is susceptible to minor deviations in positional accuracy across tool changes. However, the print strategies described in this work do not require the same sub-millimeter precision as conventional tool changing 3D printers, and clearance accounts for these minor positional deviations. The tool changing itself has a reliability of 99.6%, with 3 unsuccessful tool changes observed across the 700 performed during the fabrication of various demonstrators in this work.

7.3 Sustainability implications of mixing materials

Sustainability in digital fabrication is a growing concern in HCI [46, 47]; multi-material fabrication approaches in particular raise concerns around separation and recycling [48]. Many untreated natural fibers are biodegradable, some PLA materials are industrially compostable under controlled conditions, and some synthetic yarns may be recyclable. However, dyes, additives, blends, and the integration of yarn with filament may limit these end-of-life options. Future work could research the implications of this material mixing and experiment with yarn-filament pairs of the same composition.

8 Conclusion

We have presented FilaLoom, an add-on to consumer-grade FDM-printers that enables automatic yarn-embedding without specialist hardware, additional electronics, or permanent printer modifications. FilaLoom is a tool changer that enables the printer to smoothly switch between printing and weaving operations, opening up a design space that combines the geometric freedom of FDM-printing with the tactile- and aesthetic qualities of yarn. Through a series of custom print strategies, we demonstrated FilaLoom's potential for fabricating a wide range of structures, including three-dimensional woven objects, interactive textiles, and soft textured prints. We hope FilaLoom serves as a catalyst, encouraging a broader community of designers and researchers to explore the possibilities of yarn-embedded printing.

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